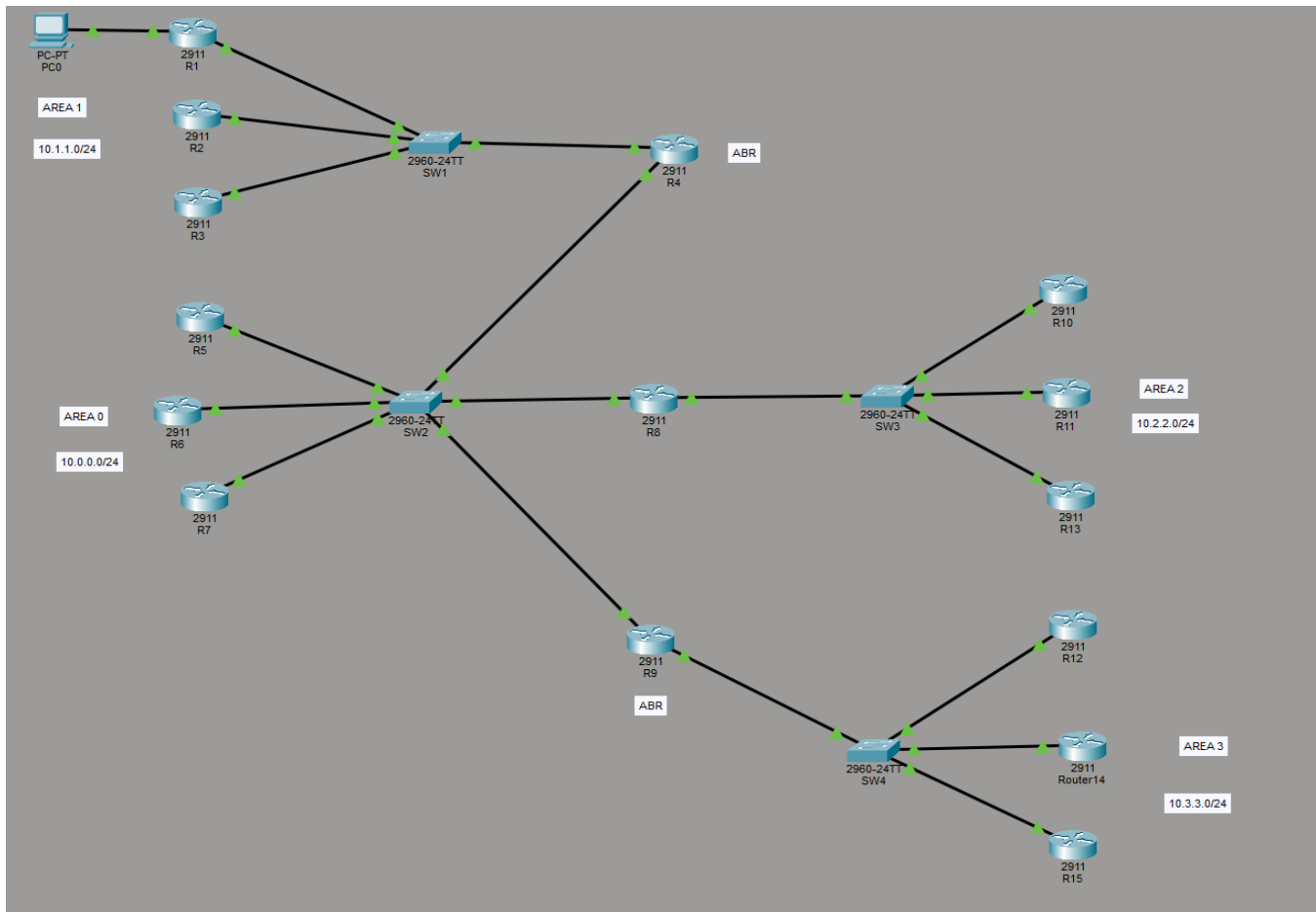


MULTI-AREA OSPF NETWORK

Backbone Area 0 with Area 1, Area 2 & Area 3 (Ethernet Topology)

DIAGRAM:



DESCRIPTION:

This project simulates a multi-area OSPF network made up of one backbone area (Area 0) and three regular areas (Area 1, Area 2 and Area 3) connected to it. Each area contains a LAN switch with a small group of routers attached, representing the end networks of that area. Two routers are configured as ABRs (Area Border Routers), each holding one interface in Area 0 and one interface in a neighboring area, which is the standard way OSPF connects multiple areas back to a single backbone. A third router links the backbone to Area 2 in the same way.

In real-world networks, large multi-area OSPF designs are used to limit the size of the link-state database on any one router, reduce SPF recalculation scope, and allow route summarization at area boundaries. This lab keeps every area small and flat, but it still demonstrates the core ideas: a single backbone area, ABRs connecting non-backbone areas to it, and inter-area routing between all four areas.

STEP 1: IP Addressing Plan (by Area):

Only the LAN network address for each area is shown below, since that is what the topology defines. Point-to-point links between switches/ABRs are not labeled in the diagram, so assign any suitable addressing (e.g. a separate /30 range) for those links when configuring.

Area	Role	Network Address	Devices on the LAN
Area 0	Backbone	10.0.0.0 /24	R5, R6, R7 (via SW2)
Area 1	Regular area	10.1.1.0 /24	R1, R2, R3, PC0 (via SW1)
Area 2	Regular area	10.2.2.0 /24	R10, R11, R13 (via SW3)
Area 3	Regular area	10.3.3.0 /24	R12, Router14, R15 (via SW4)

STEP 2: Configure Area 1 — SW1, R1, R2, R3 & PC0:

SW1 simply needs each router-facing port set as an access port (no VLANs required, since all three routers and PC0 sit on the same Area 1 LAN), plus the uplink port toward the ABR.

SW1 (2960):

```
interface range fa0/1 - 3 switchport mode access
```

```
interface fa0/24
```

```
switchport mode trunk (or leave as a normal access/uplink port if no VLANs are used)
```

R1, R2, R3 (each):

```
interface Gig0/0
```

```
ip address <assign from 10.1.1.0/24> no shutdown
```

```
router ospf 1
```

```
network 10.1.1.0 0.0.0.255 area 1
```

PC0 hangs directly off R1 (Gig0/1) and represents an end host inside Area 1 — give it an address from 192.168.1.0/24 with R1's interface as its default gateway.

STEP 3: Configure the Area 0 Backbone — SW2, R5, R6, R7:

SW2 (2960-24TT):

```
interface range fa0/1 - 3 switchport
```

```
mode access
```

R5, R6, R7 (each):

```
interface Gig0/0
```

```
ip address <assign from 10.0.0.0/24> no shutdown
```

```
router ospf 1
```

```
network 10.0.0.0 0.0.0.255 area 0
```

STEP 4: Configure ABR1 (Area 1 <-> Area 0):

This router has one interface toward SW1 (Area 1 side) and one interface toward SW2 (Area 0 side). The interface facing each switch is placed into the OSPF area that side actually belongs to — that is what makes it an ABR.

ABR1:

```
interface Gig0/0 (toward SW1 / Area 1) ip address <point-to-point address>
no shutdown
interface Gig0/1 (toward SW2 / Area 0) ip address <point-to-point address>
no shutdown
router ospf 1
network <Gig0/0 link network> area 1 network <Gig0/1 link network> area 0
```

STEP 5: Configure the Link to Area 2 — R8, SW3, R10, R11, R13:

R8 connects the Area 0 backbone (via SW2) to the Area 2 switch (SW3), so it carries the same Area 0 / Area 2 split on its two interfaces as the dedicated ABRs do.

SW3 (2960-24TT):

```
interface range fa0/1 - 3 switchport mode access
```

R8:

```
interface Gig0/0 (toward SW2 / Area 0) ip address <point-to-point address>
no shutdown
interface Gig0/1 (toward SW3 / Area 2) ip address <point-to-point address>
no shutdown
router ospf 1
network <Gig0/0 link network> area 0 network <Gig0/1 link network> area 2
```

R10, R11, R13 (each):

```
interface Gig0/0
ip address <assign from 10.2.2.0/24> no shutdown
router ospf 1
network 10.2.2.0 0.0.0.255 area 2
```

STEP 6: Configure ABR2 (Area 0 <-> Area 3):

ABR2:

```
interface Gig0/0 (toward SW2 / Area 0) ip address <point-to-point address>
no shutdown
interface Gig0/1 (toward SW4 / Area 3) ip address <point-to-point address>
no shutdown
router ospf 1
network <Gig0/0 link network> area 0 network <Gig0/1 link network> area 3
```

STEP 7: Configure Area 3 — SW4, R12, Router14, R15:

SW4 (2960-24TT):

```
interface range fa0/1 - 3 switchport mode access
```

R12, Router14, R15 (each):

```
interface Gig0/0
ip address <assign from 10.3.3.0/24> no shutdown
router ospf 1
network 10.3.3.0 0.0.0.255 area 3
```

STEP 8: OSPF Area Assignment Summary:

Router	Role	Area(s) on its interfaces
R1, R2, R3	Area 1 LAN routers	Area 1
R5, R6, R7	Area 0 (backbone) LAN routers	Area 0
R10, R11, R13	Area 2 LAN routers	Area 2
R12, Router14, R15	Area 3 LAN routers	Area 3
ABR1	Area 1 <-> Area 0 border	Area 1 and Area 0
R8	Area 0 <-> Area 2 border	Area 0 and Area 2
ABR2	Area 0 <-> Area 3 border	Area 0 and Area 3

STEP 9: Verification:

Run on the routers/ABRs as needed:

show ip route

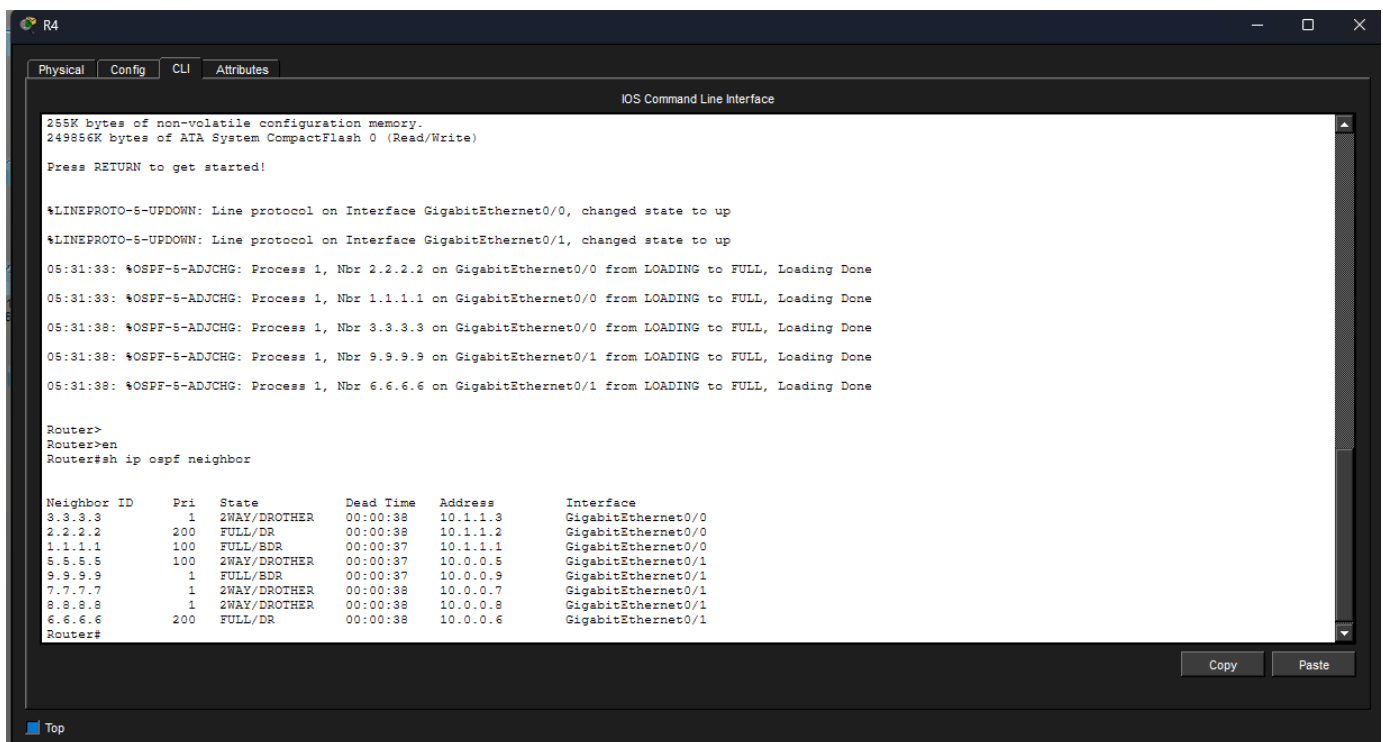
show ip route ospf show ip ospf neighbor

show ip ospf interface brief show ip ospf database

show ip protocols

show ip interface brief

show ip ospf neighbor (ABR1 or ABR2):



The screenshot shows a Cisco IOS Command Line Interface window titled "R4". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area displays the following text:

```
165K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
05:31:33: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
05:31:33: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
05:31:38: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
05:31:38: %OSPF-5-ADJCHG: Process 1, Nbr 9.9.9.9 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
05:31:38: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

Router>
Router>en
Router#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	2WAY/DROTHER	00:00:38	10.1.1.3	GigabitEthernet0/0
2.2.2.2	200	FULL/DR	00:00:38	10.1.1.2	GigabitEthernet0/0
1.1.1.1	100	FULL/BDR	00:00:37	10.1.1.1	GigabitEthernet0/0
5.5.5.5	100	2WAY/DROTHER	00:00:37	10.0.0.5	GigabitEthernet0/1
9.9.9.9	1	FULL/BDR	00:00:37	10.0.0.9	GigabitEthernet0/1
7.7.7.7	1	2WAY/DROTHER	00:00:38	10.0.0.7	GigabitEthernet0/1
8.8.8.8	1	2WAY/DROTHER	00:00:38	10.0.0.8	GigabitEthernet0/1
6.6.6.6	200	FULL/DR	00:00:38	10.0.0.6	GigabitEthernet0/1

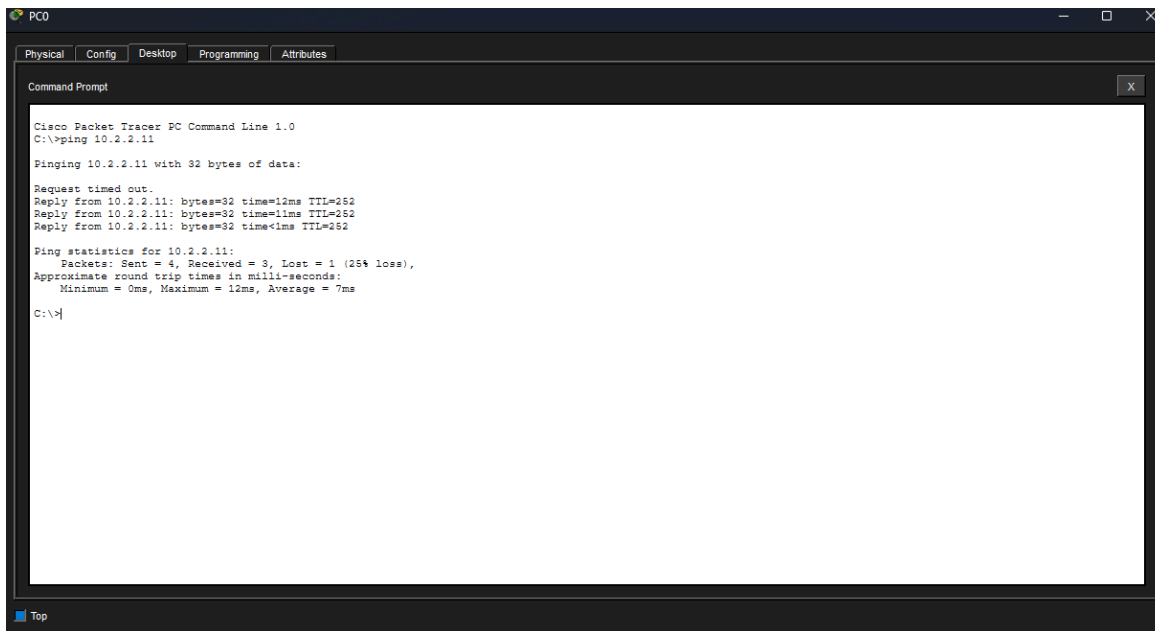
Router#

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

STEP 10: Ping Test:

Verify end-to-end, , for example from PC0 (Area 1) to a host in Area 2 and a host in Area 3.

Ping from PC0 (Area 1) to a host in Area 2 (10.2.2.0/24):



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.2.2.11

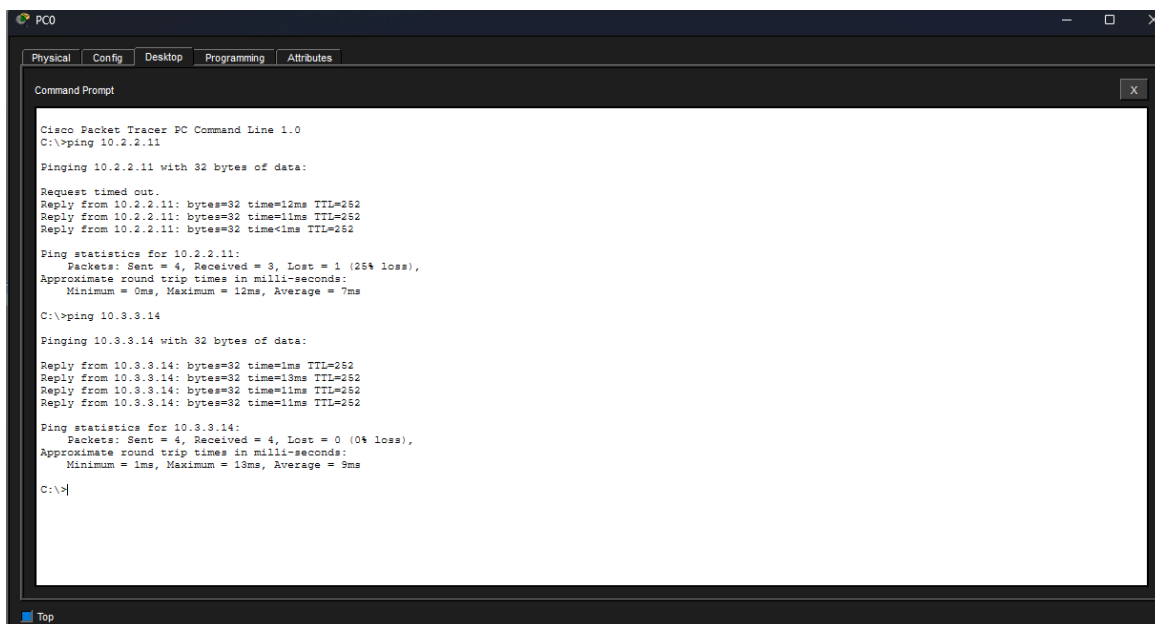
Pinging 10.2.2.11 with 32 bytes of data:

Request timed out.
Reply from 10.2.2.11: bytes=32 time=12ms TTL=252
Reply from 10.2.2.11: bytes=32 time=11ms TTL=252
Reply from 10.2.2.11: bytes=32 time<1ms TTL=252

Ping statistics for 10.2.2.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 7ms

C:\>
```

Ping from PC0 (Area 1) to a host in Area 3 (10.3.3.0/24):



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.2.2.11

Pinging 10.2.2.11 with 32 bytes of data:

Request timed out.
Reply from 10.2.2.11: bytes=32 time=12ms TTL=252
Reply from 10.2.2.11: bytes=32 time=11ms TTL=252
Reply from 10.2.2.11: bytes=32 time<1ms TTL=252

Ping statistics for 10.2.2.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 7ms

C:\>ping 10.3.3.14

Pinging 10.3.3.14 with 32 bytes of data:

Reply from 10.3.3.14: bytes=32 time=1ms TTL=252
Reply from 10.3.3.14: bytes=32 time=13ms TTL=252
Reply from 10.3.3.14: bytes=32 time=11ms TTL=252
Reply from 10.3.3.14: bytes=32 time=11ms TTL=252

Ping statistics for 10.3.3.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 9ms

C:\>
```

NOTE: All four areas (0, 1, 2 and 3) sit on the same internal LAN in this lab for simplicity — the focus here is on multi-area OSPF and ABR behavior rather than WAN transport.